



# Computer Vision

## Camera Geometry

Professor Jeong, Mun-Ho

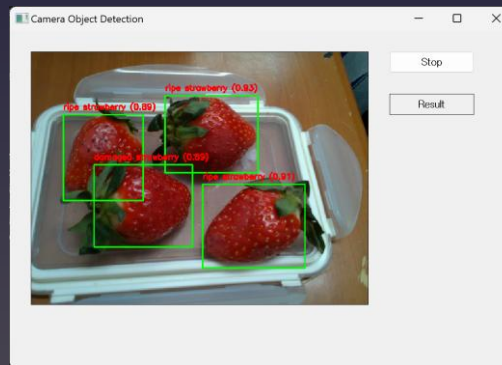
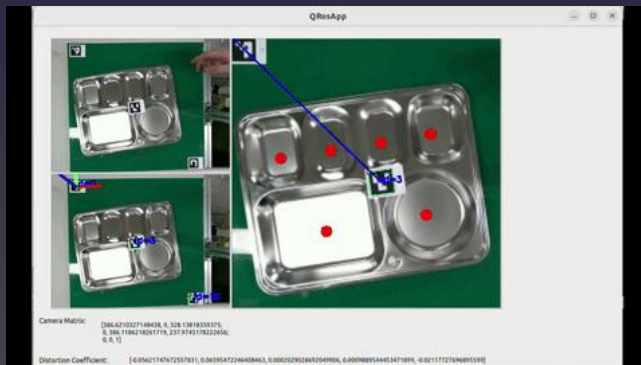
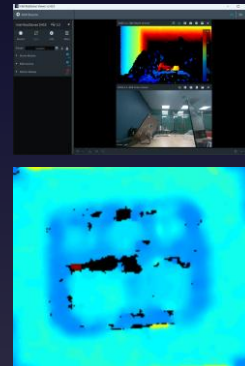
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# Schedule

| Week                                                                                | Topics                       |                                                |                                               | HW | Quiz |
|-------------------------------------------------------------------------------------|------------------------------|------------------------------------------------|-----------------------------------------------|----|------|
| 1                                                                                   | Introduction                 |                                                | Introduction(1,9/2)                           |    |      |
|  2 | Fundamental Image Processing | Camera Geometry(2, 9/7)                        | Color Transform(3, 9/9)<br>Basics(4. 9/9)     | 1  | 1    |
| 3                                                                                   |                              | Contrast, Image Thresholding<br>Contrast       | Otsu's Algorithm<br>Image Histogram           |    |      |
| 4                                                                                   |                              | Image Thresholding                             | Convolution, Box Filtering                    |    |      |
| 5                                                                                   |                              | Gaussian / Median Filtering<br>Image Filtering | Histogram Transform<br>Histogram Equalization |    |      |
| 6                                                                                   |                              | Image Subsampling                              | Edge Extraction                               |    |      |
| 7                                                                                   | Feature Extraction           | Canny Operator, Edge with FDG                  | Corner Detection, Harris Corner               |    |      |
| 8                                                                                   |                              | Mid-term Exam.                                 |                                               |    |      |
| 9                                                                                   |                              | We-Meet & IC-PBL Mentor Presentation           | Hough Transform                               |    |      |
| 10                                                                                  |                              | Generalized Hough T., Circle Hough T.          |                                               |    |      |
| 11                                                                                  |                              | ArUCo Marker Detection                         | ArUCo Marker Detection                        |    |      |
| 12                                                                                  |                              | Optical Flow                                   | Optical Flow                                  |    |      |
| 13                                                                                  |                              | Optical Flow, ImagePyramid                     | We-Meet: IC-PBL                               |    |      |
| 14                                                                                  |                              |                                                |                                               |    |      |
| 15                                                                                  |                              |                                                |                                               |    |      |
| 16                                                                                  |                              | We-Meet: IC-PBL                                | Final Exam                                    |    |      |

# IC-PBL

- Problem-based Learning
- Problems from an Industry Mentor



# Today's Topics

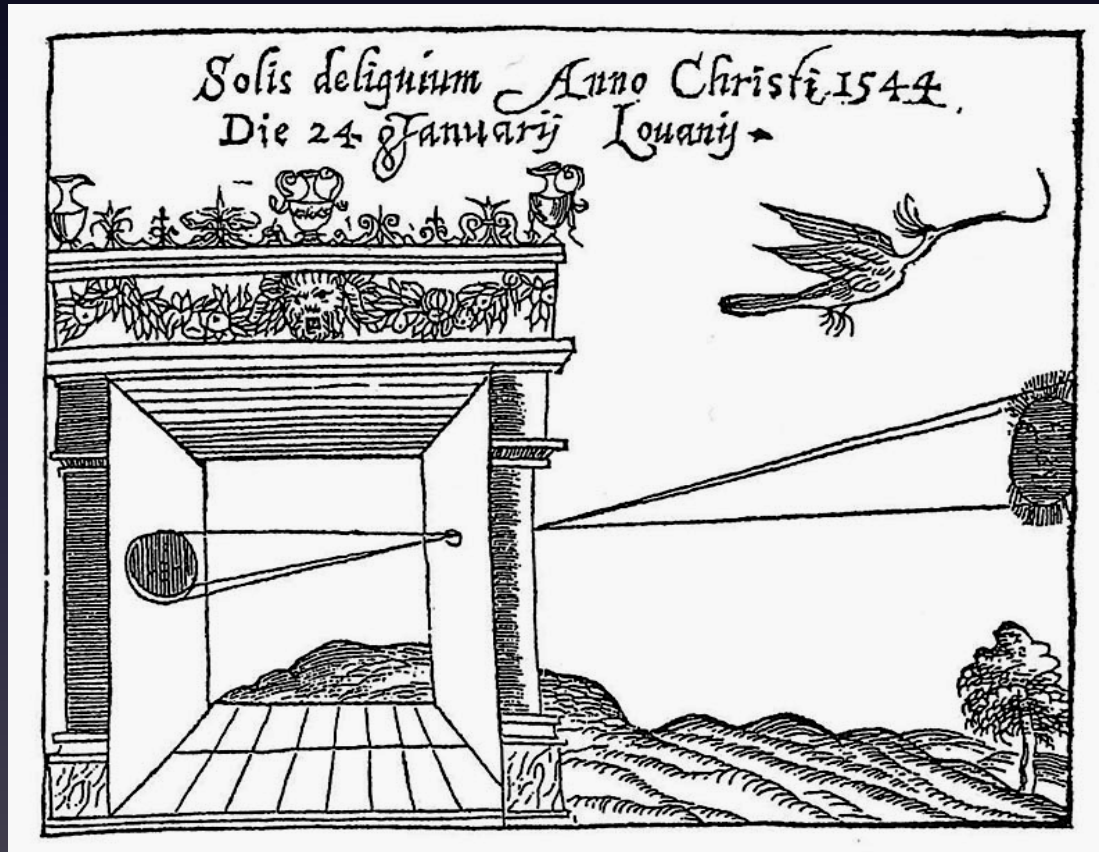
- Camera Geometry

# Camera Geometry

- Brief History of Camera
- Pin-Hole Camera
- Use of Lenses
- Perspective Projection
- Lens Distortion

# Brief History of Camera

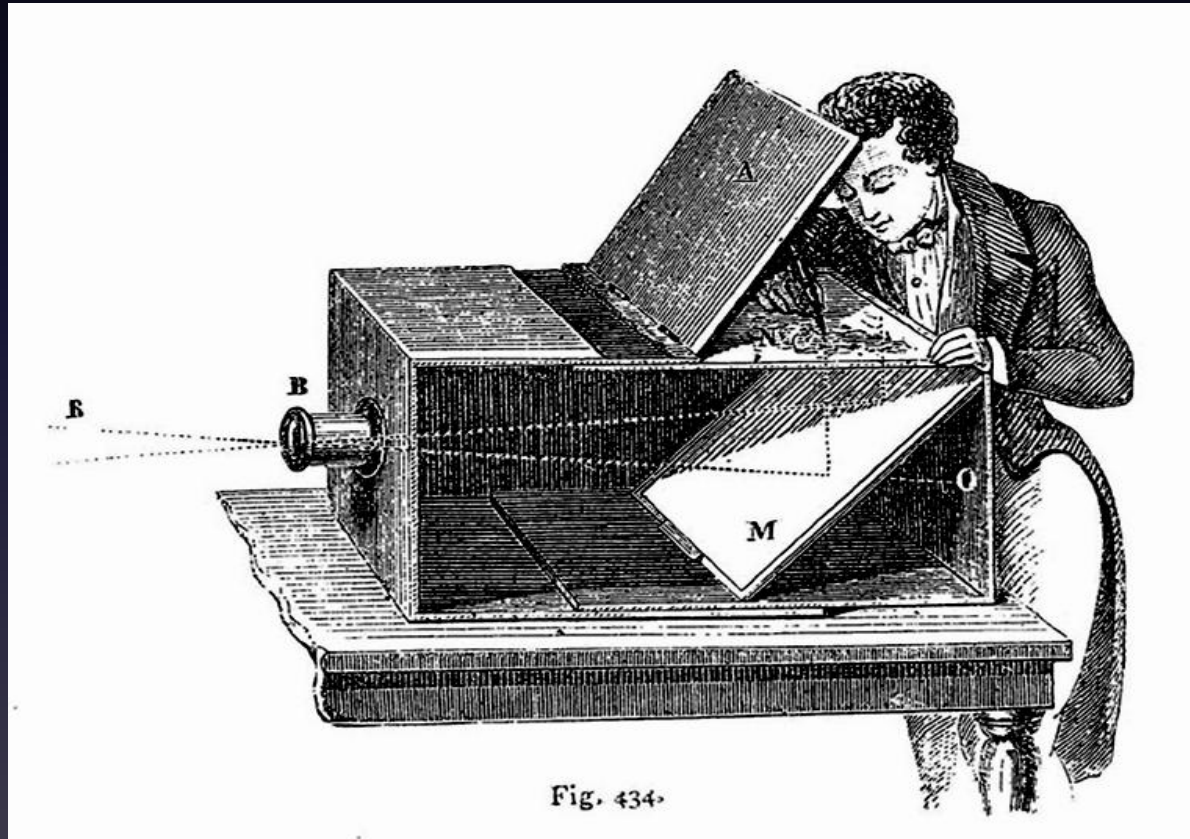
- Camera obscura



Camera Obscura, Gemma Frisius, 1544



# Lens-Based Camera Obscura



Lens Based Camera Obscura, 1568

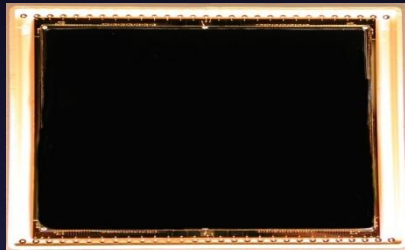
# Lens-Based Camera Obscura



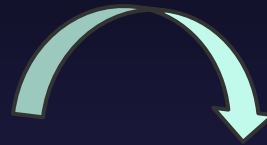
<https://www.flickr.com/photos/randomblue/3860374904/>



# CCD/CMOS Camera



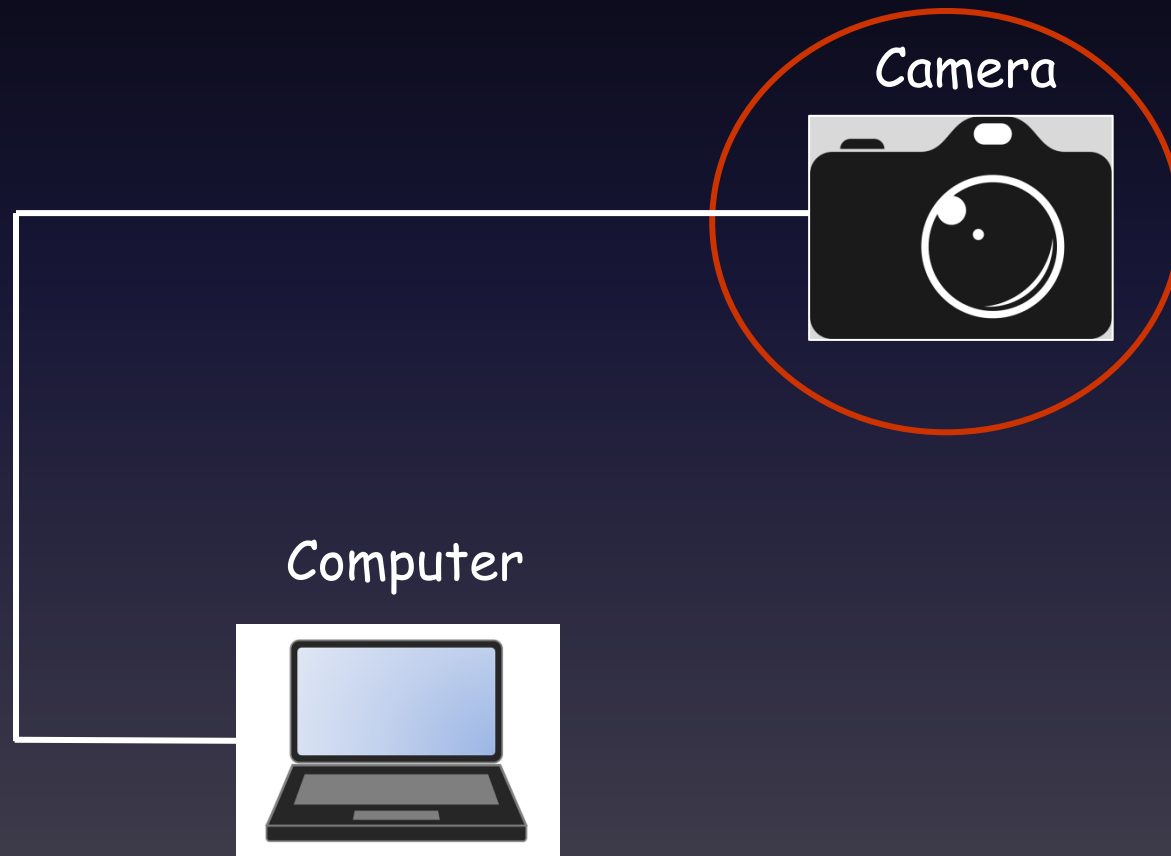
Silicon Image Detector, 1970



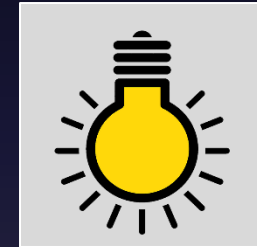
Digital Cameras



# Computer Vision System



Lighting

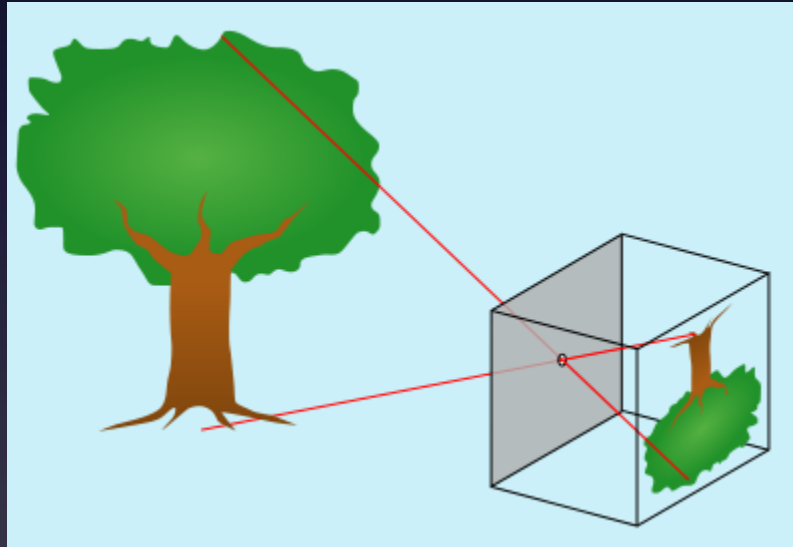


Scene



# Pinhole Camera

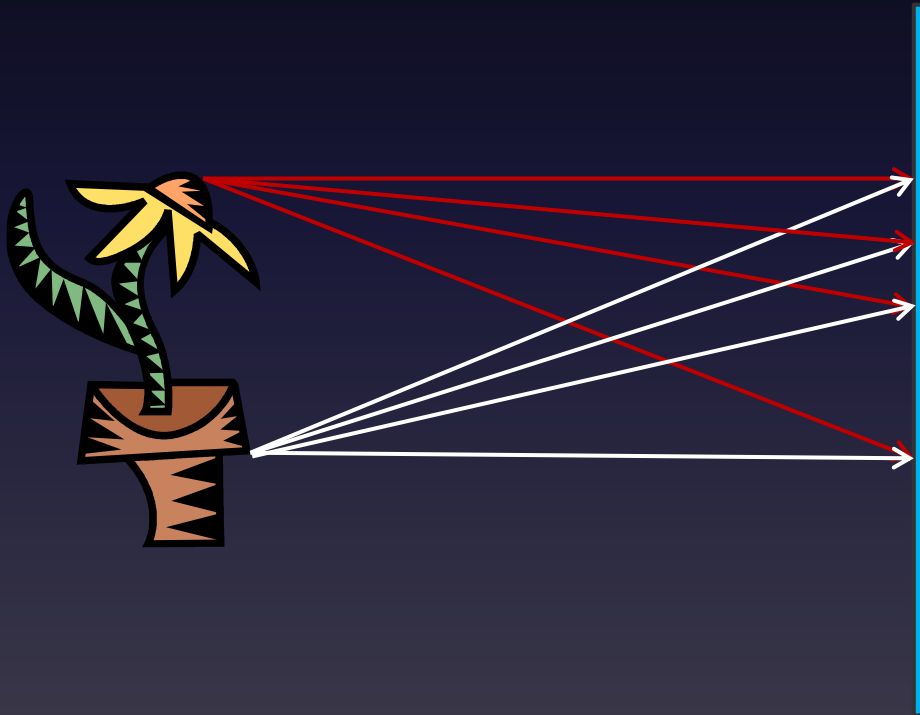
- Dark box with a small hole and photographic paper
- Creates an inverted image on the photographic paper



<https://commons.wikimedia.org/wiki/File:Pinhole-camera.svg>

# Why Pinhole Camera?

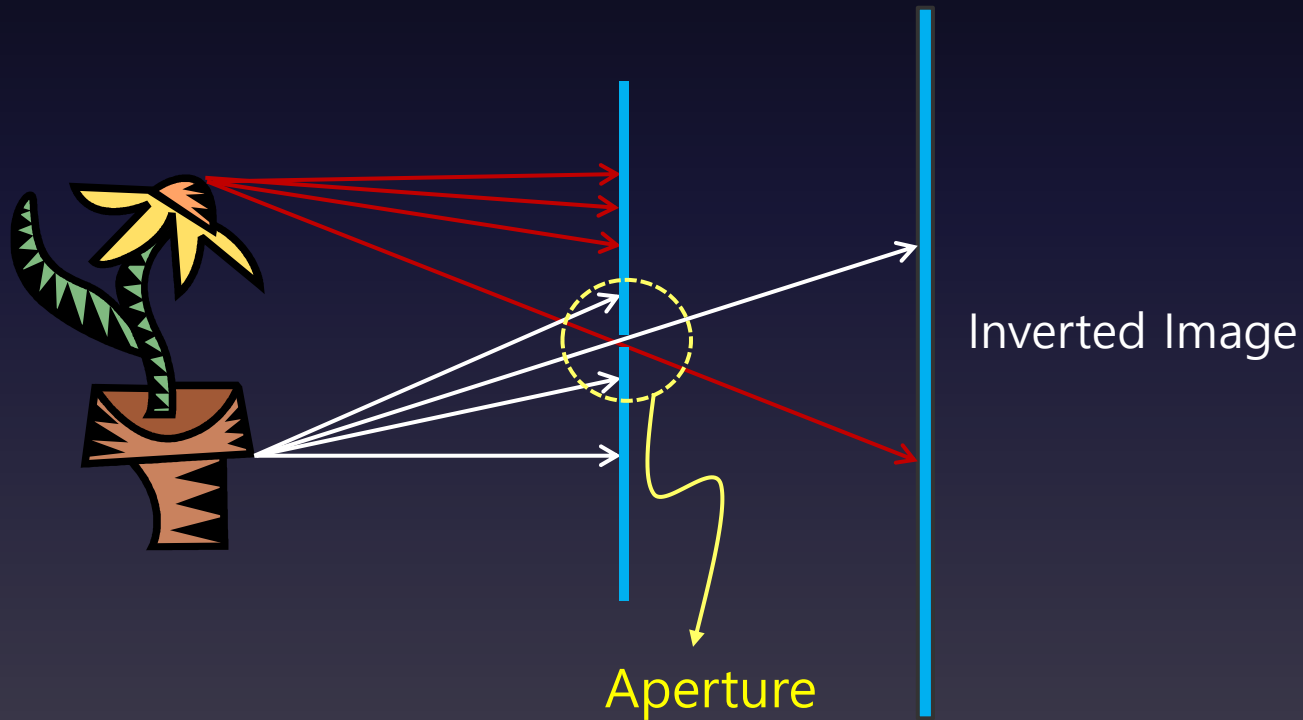
- Try to Design a Camera !



*Need to block off most of rays !*

# Why Pinhole Camera?

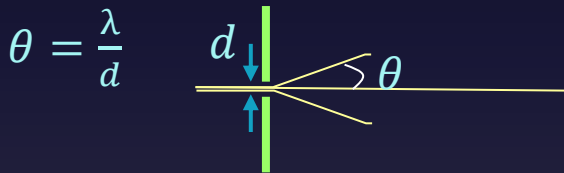
- Try to Design a Camera !



*This reduces Blurring, but too Dark !*

# Problems with Pinholes

- A narrow pinhole size
  - better the focus
  - less light received by image plane.
  - diffraction blurring the image



- The focus is best when

$$d = 2 \sqrt{f' \lambda}.$$



**Fig. 5.96** The pinhole camera. Note the variation in image clarity as the hole diameter decreases. [Photos courtesy Dr. N. Joel, UNESCO.]



# Pinhole Photography

- Image Size inversely proportional to Distance



<https://www.flickr.com/photos/hartman045/4445819542>



<http://www.pinholerresource.com/>

# Pinhole Photography



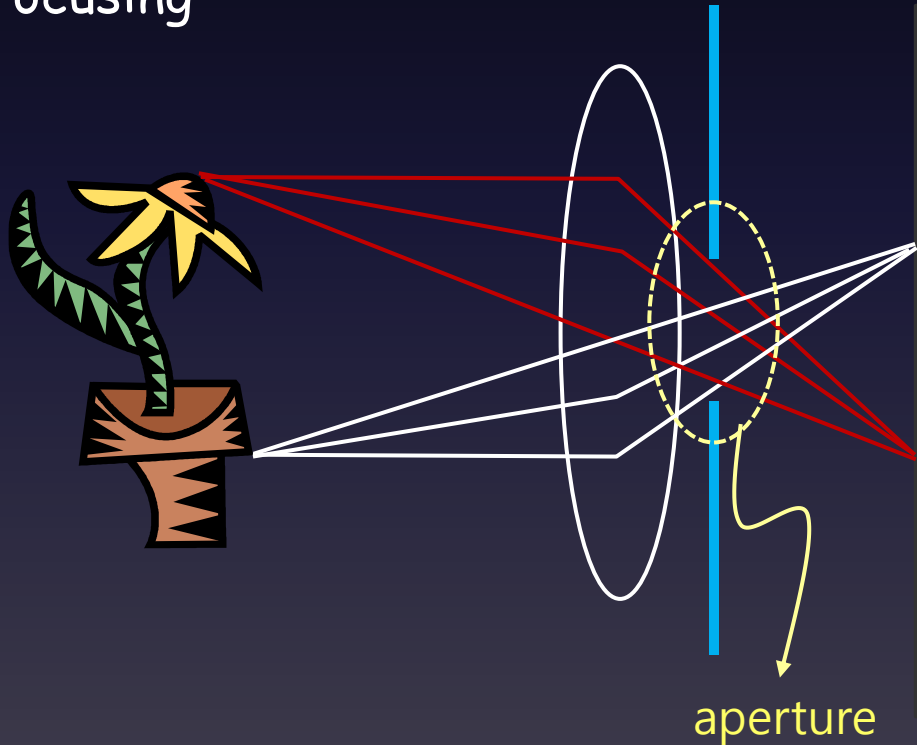
Exposure 4 seconds



Exposure 96 minutes

# Use of Lenses

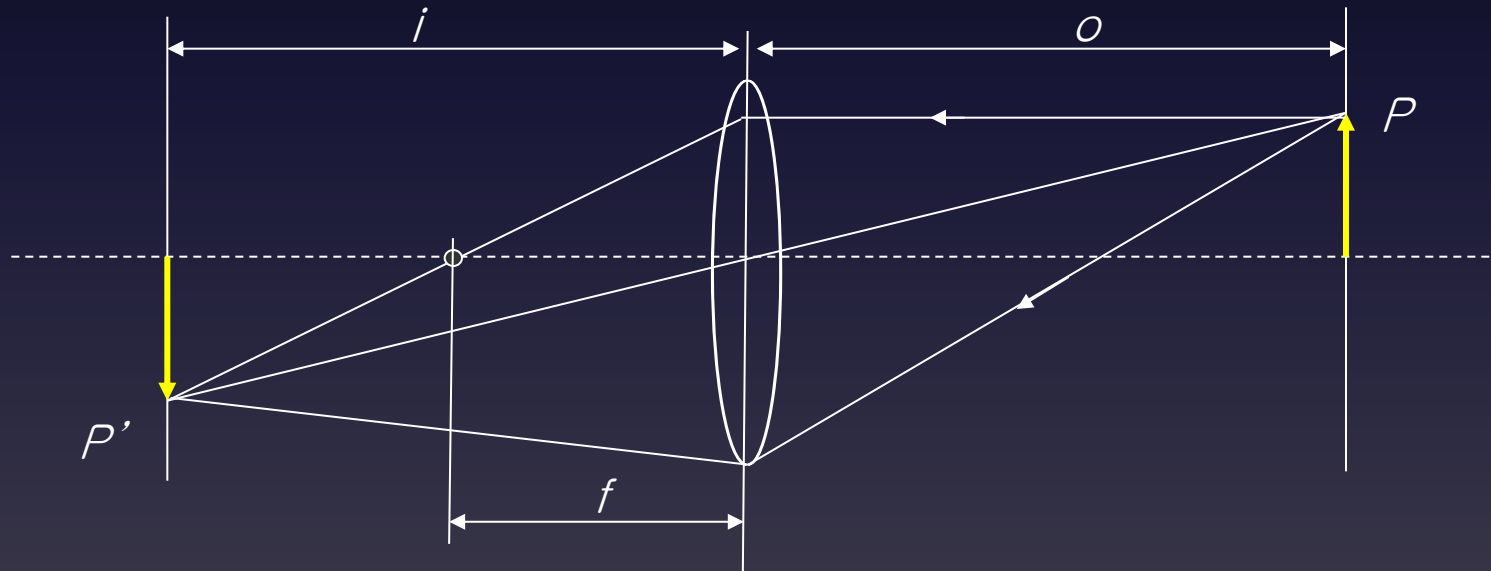
- Gathering more light
- Sharp focusing



*Still obey the Pinhole Projection.*

# Thin Lens Formula

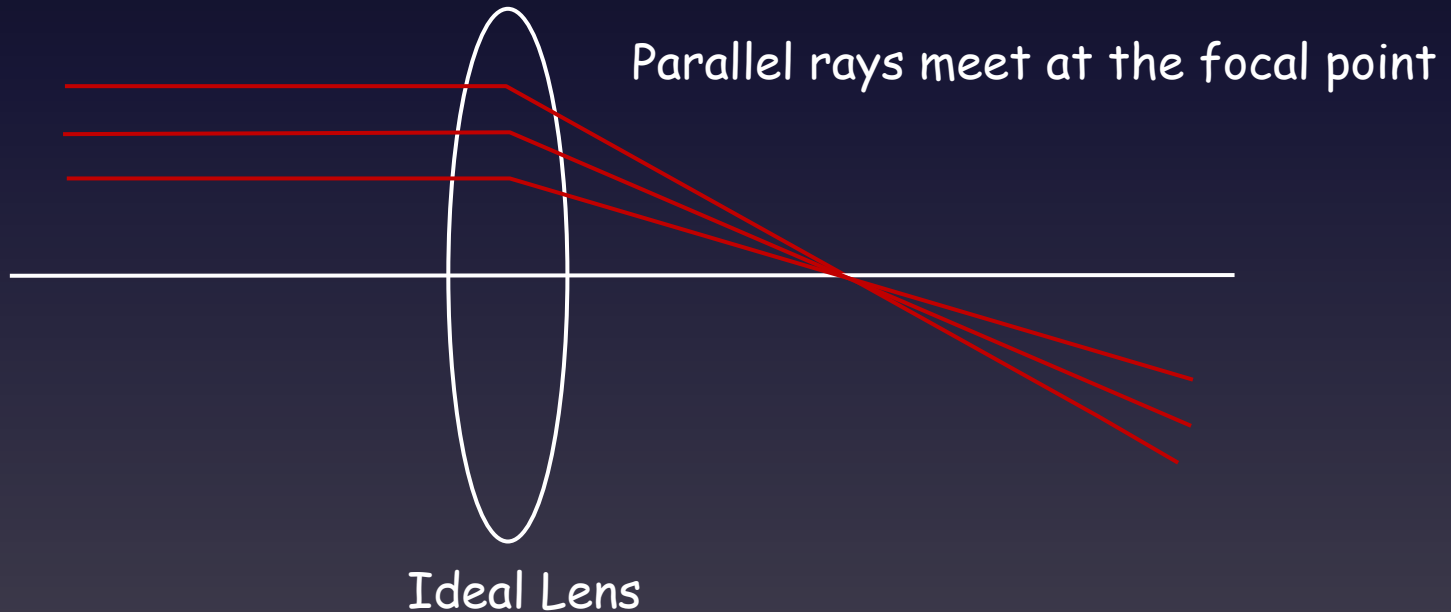
- Lenses are used to avoid problems with pinholes.
- Ideal Lens: Same projection as pinhole but gathers more lights!



- Gaussian Thin Lens Formula:  $\frac{1}{i} + \frac{1}{o} = \frac{1}{f}$
- $f$  is the focal length of the lens – determines the lens's ability to refract light

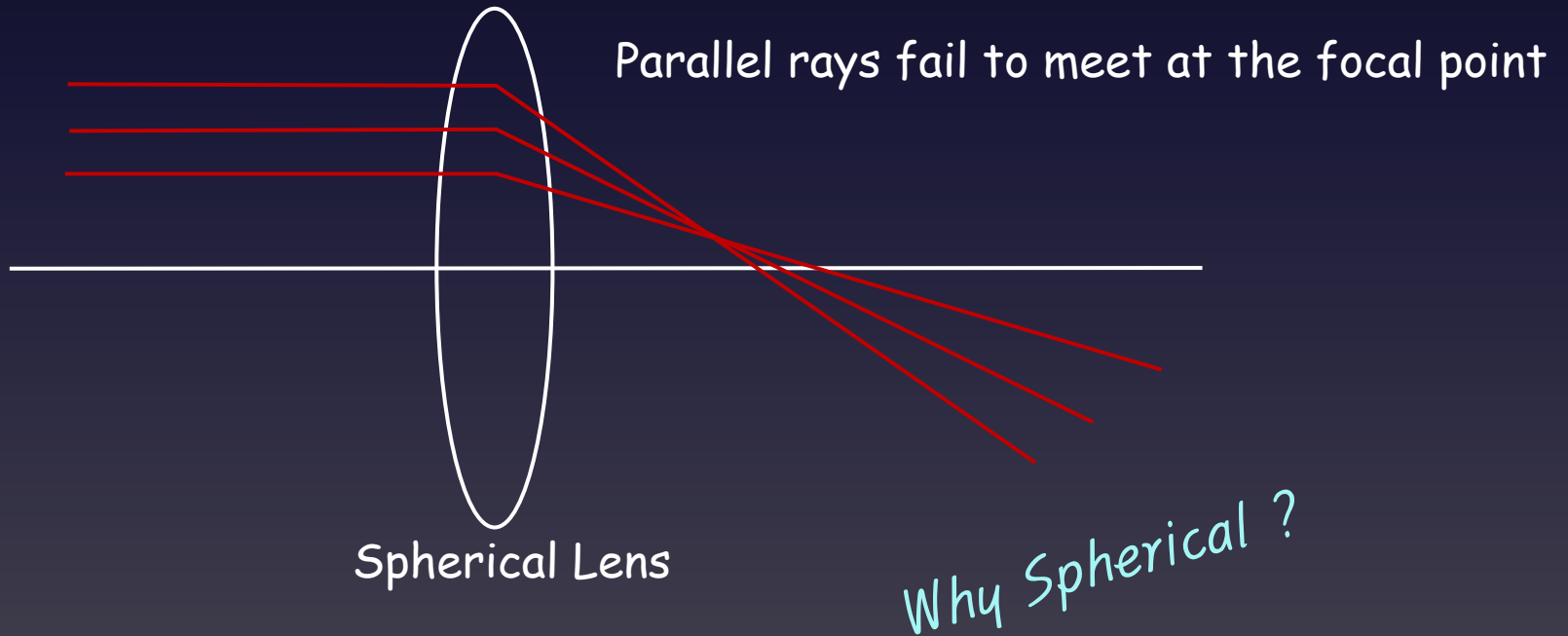
# Lens Caused Problems

- Spherical Aberration



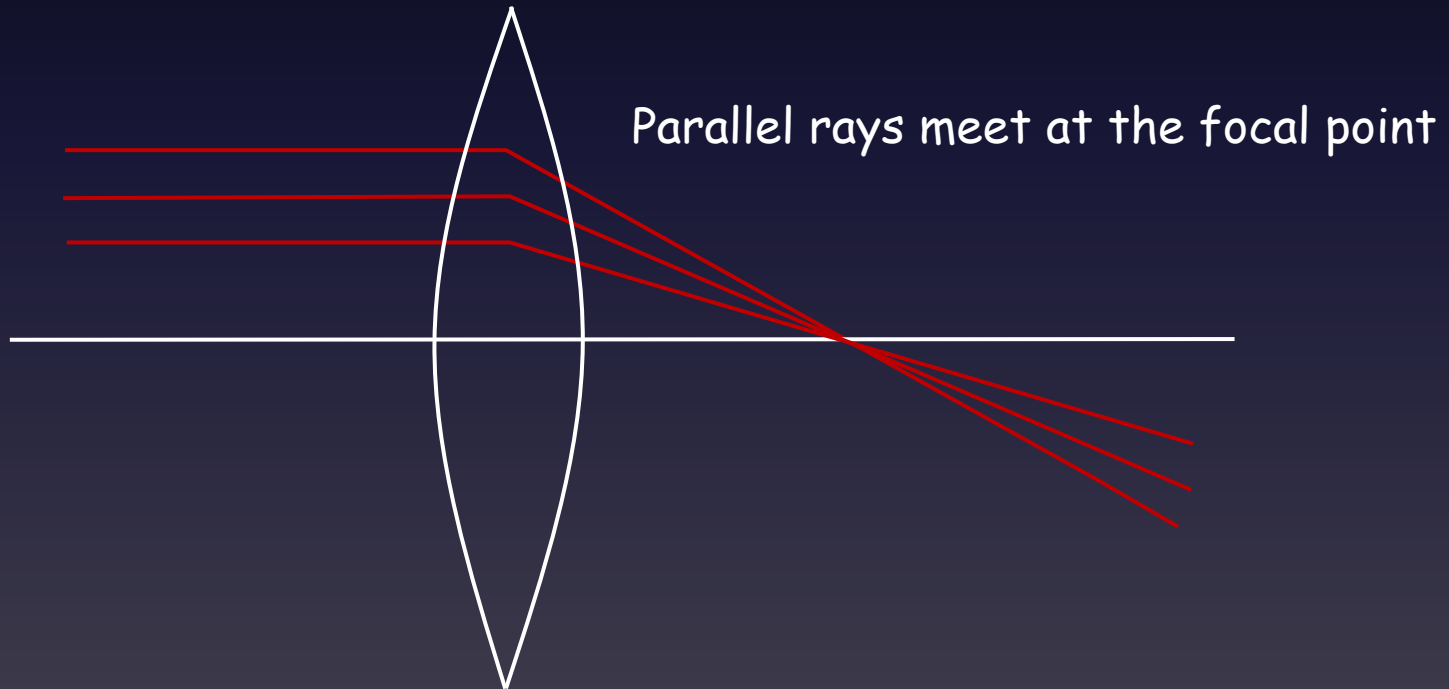
# Lens Caused Problems

- Spherical Aberration



# To Reduce Spherical Aberration

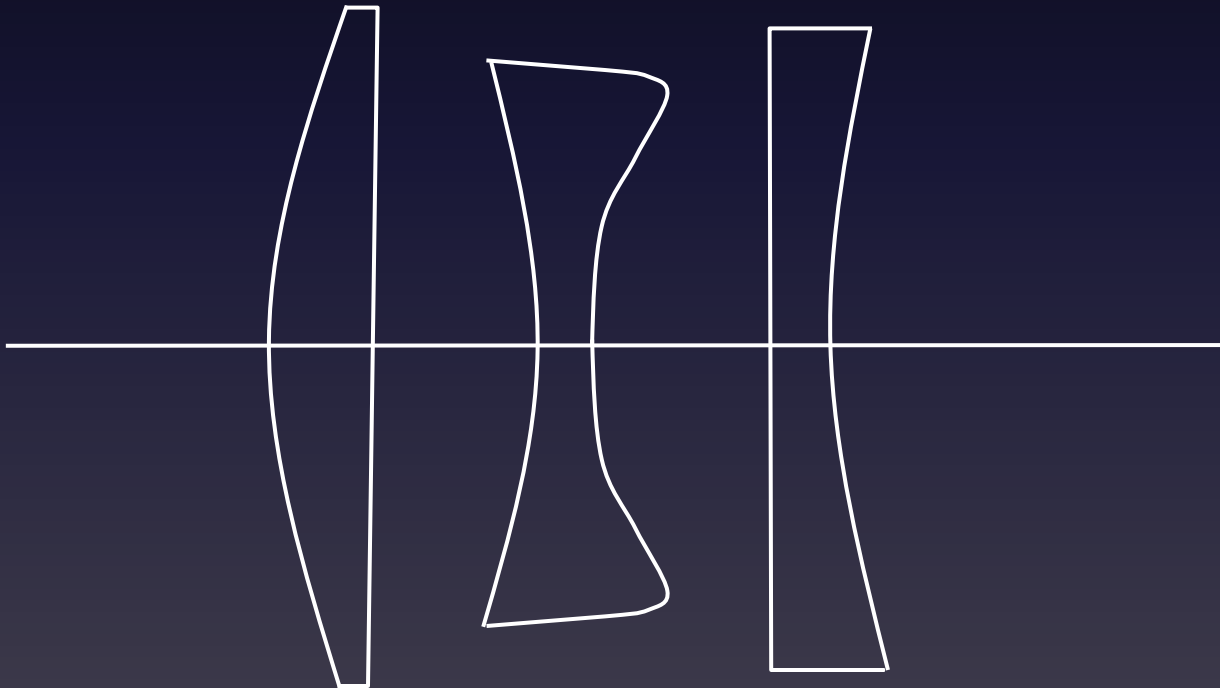
- Non-Spherical Lens





# To Reduce Spherical Aberration

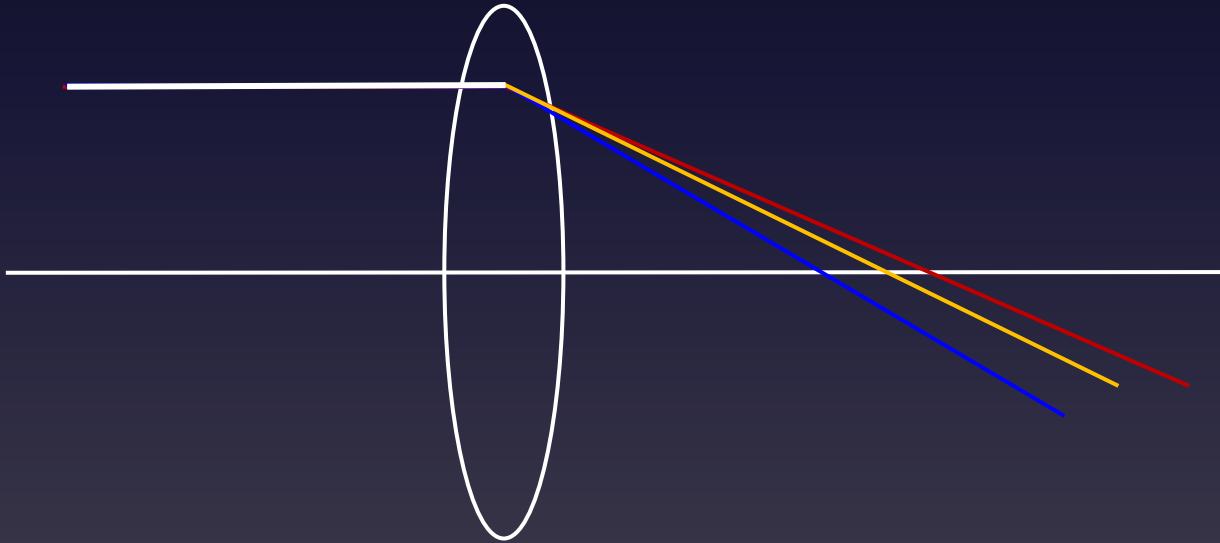
- Multiple Lens



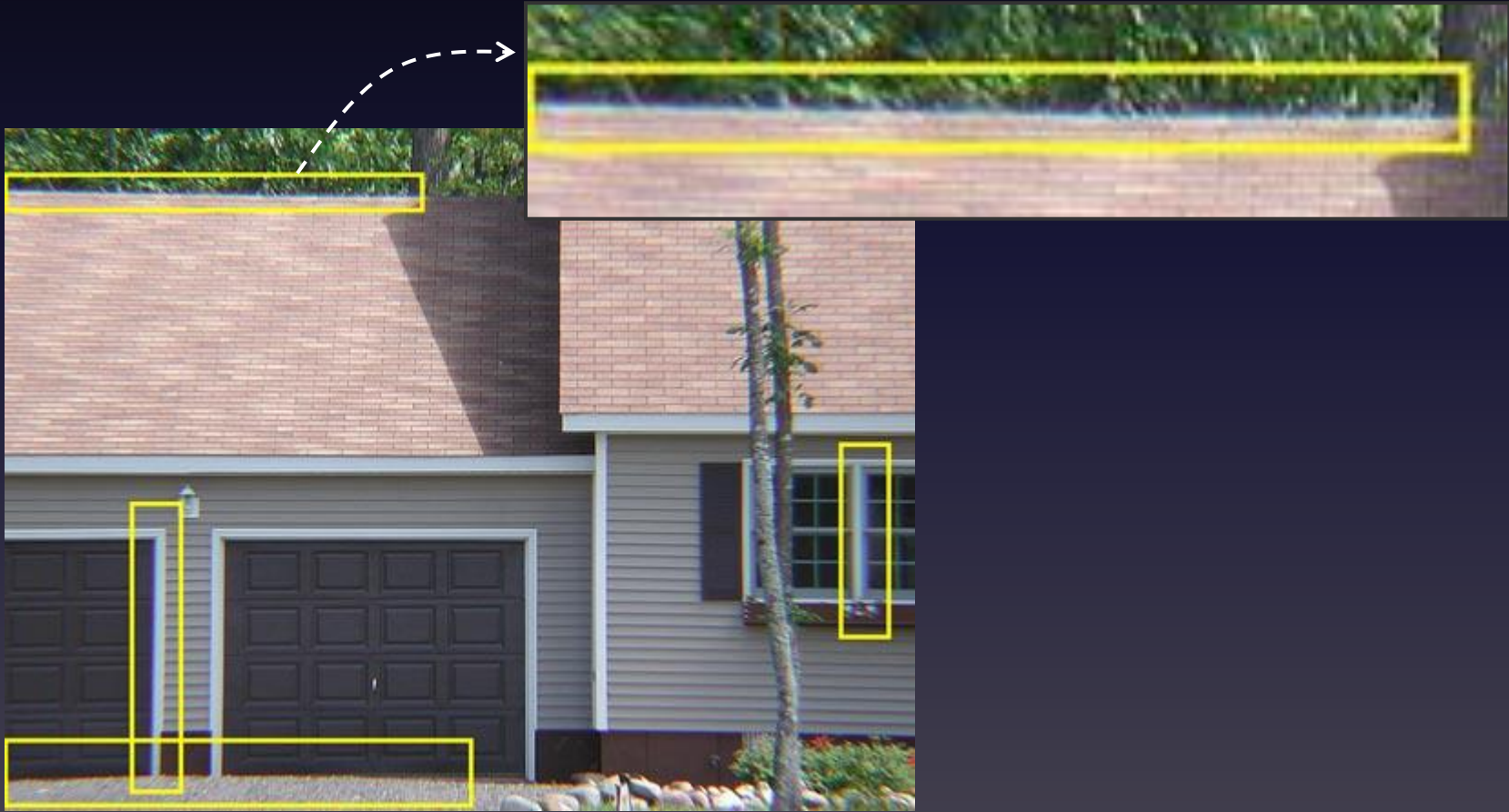
# Lens Caused Problems

- Chromatic Aberration

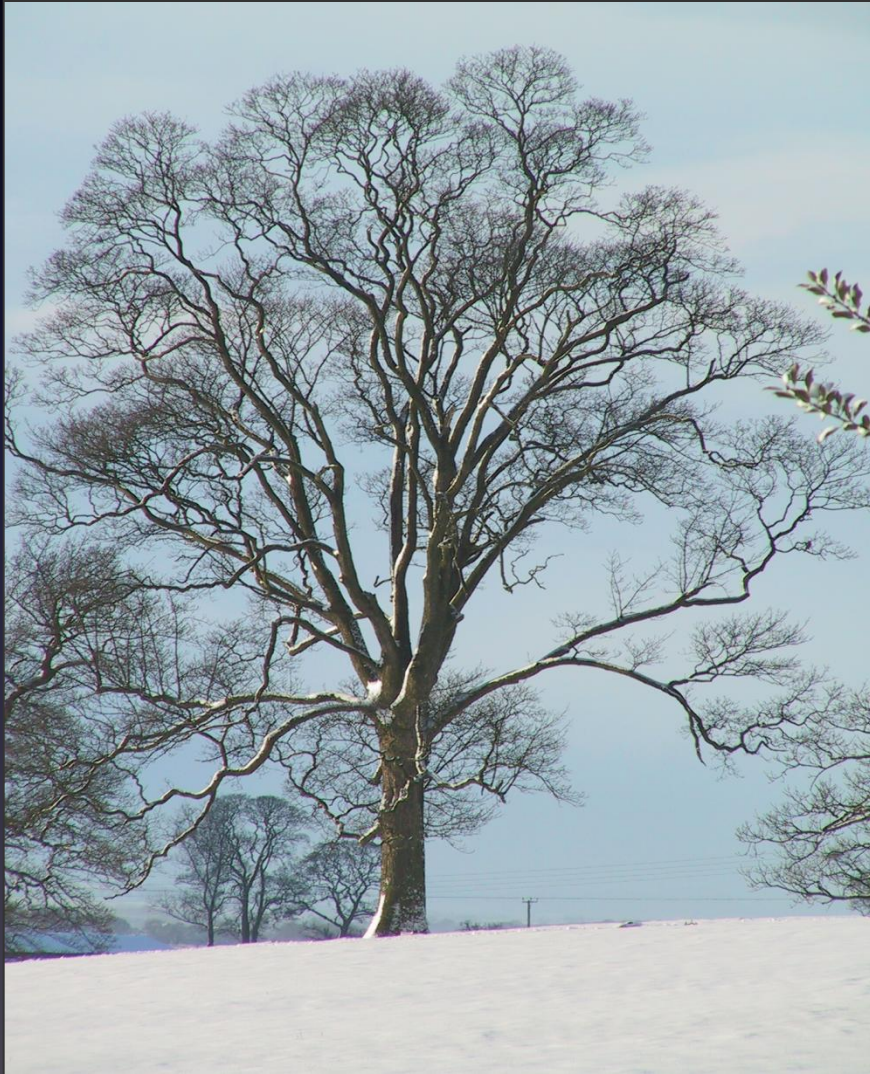
- Due to different refractive indices for different wavelengths



# Chromatic Aberration



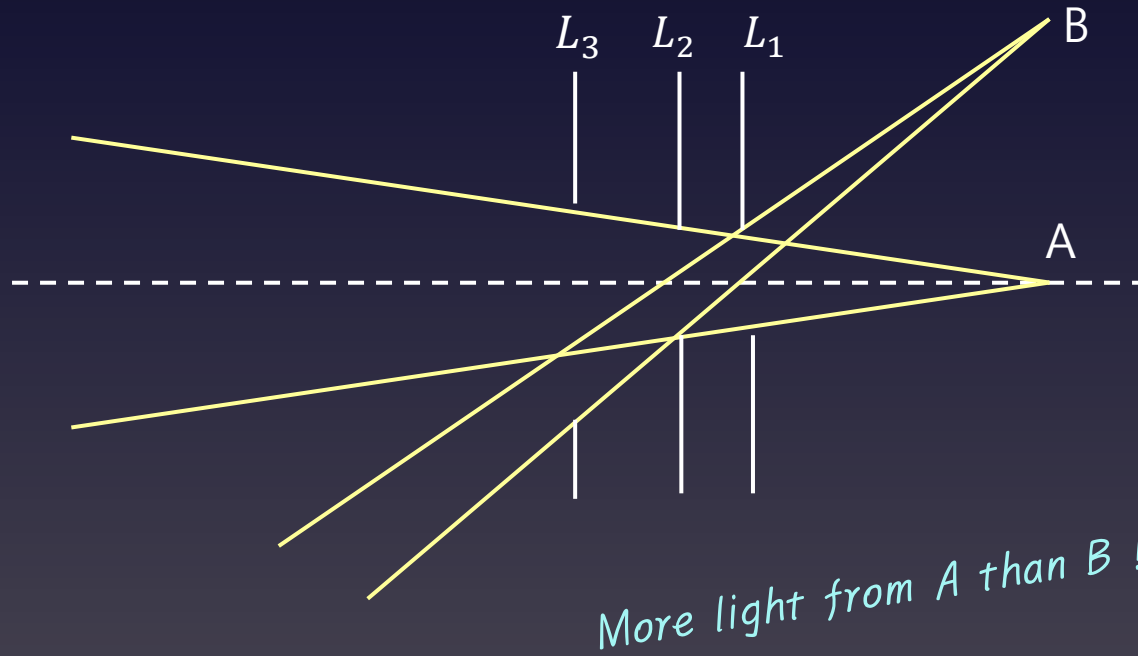
# Chromatic Aberration



# Lens Caused Problems

## ■ Vignette

- Light rays from a point are partially blocked by the various apertures of the multiple lenses



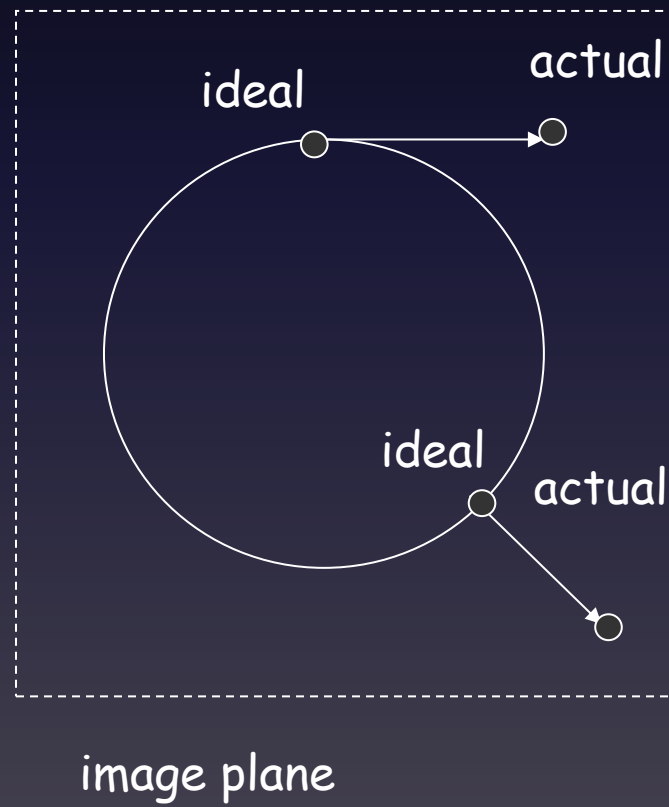


# Vignette



# Lens Caused Problems

- Radial Distortion and Tangential Distortion



# Lens Caused Problems

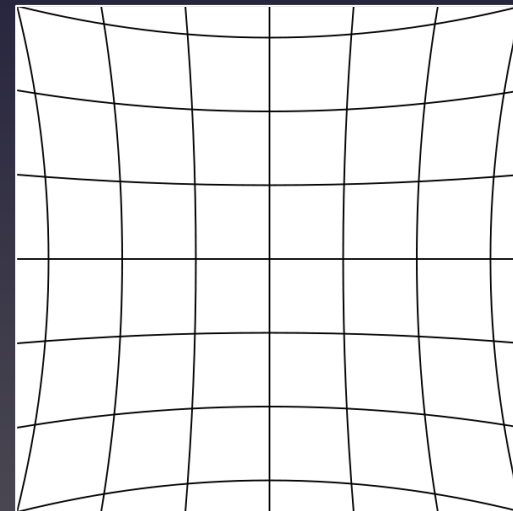
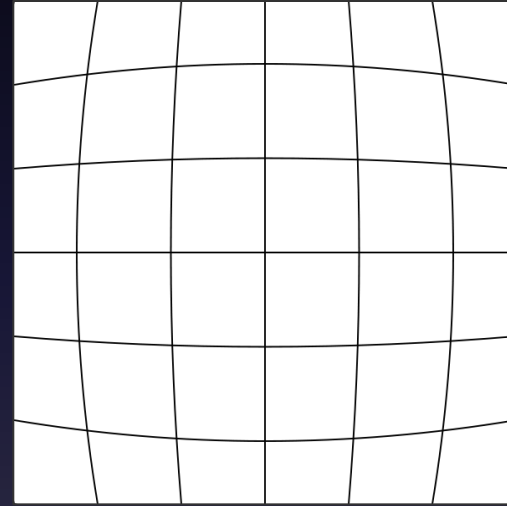
- Radial Distortion

- Barrel distortion

Image magnification decreases with distance from the optical axis

- Pincushion distortion

Image magnification increases with the distance from the optical axis

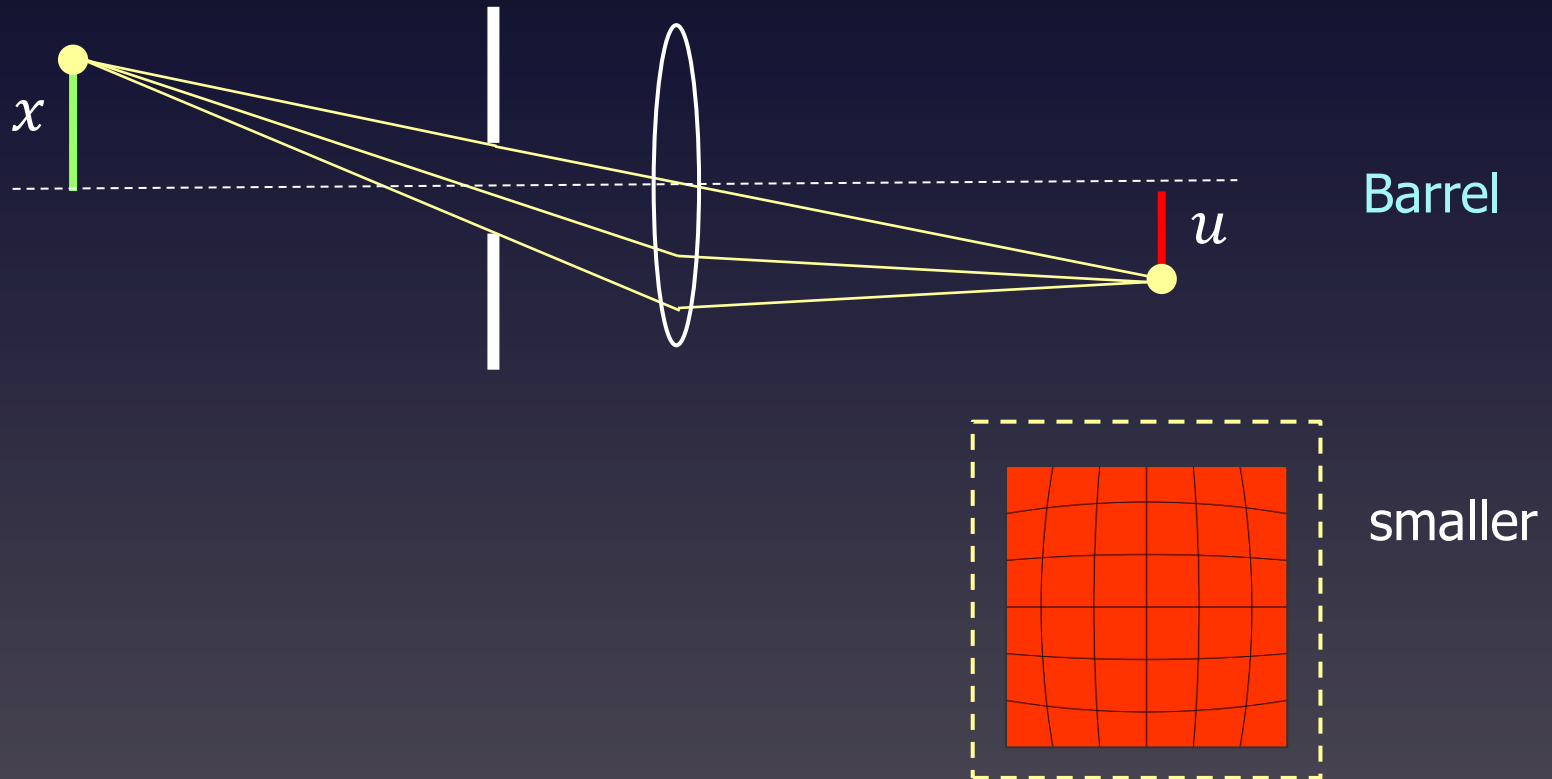




# Lens Caused Problems

## ■ Cause of Distortion

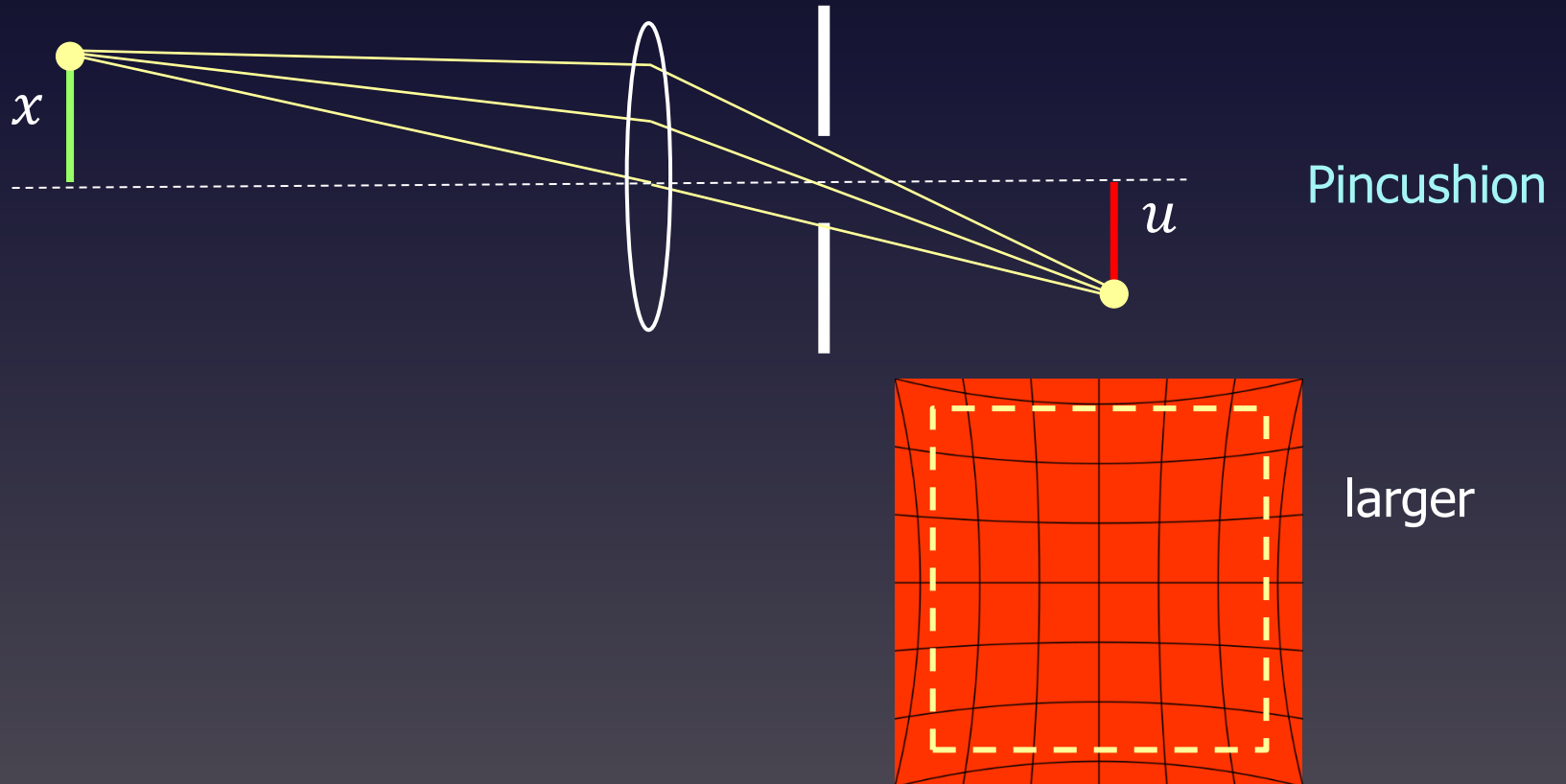
- Introduction of a stop in front
- Common in wide-angle lenses



# Lens Caused Problems

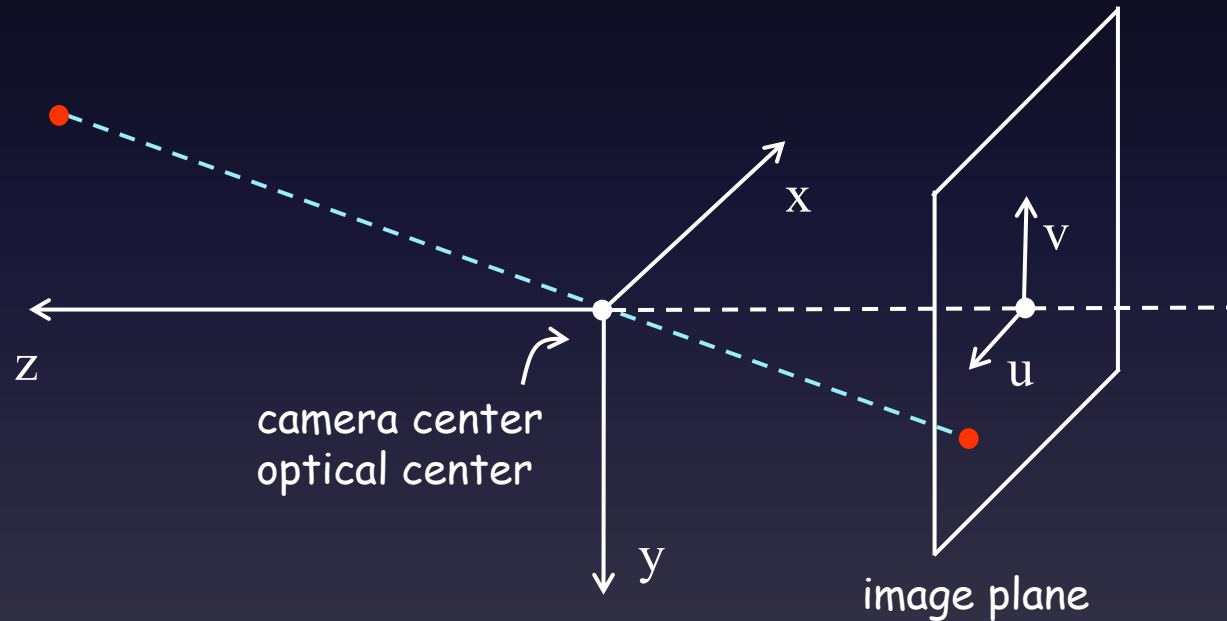
## ■ Cause of Distortion

- Introduction of a stop in rear
- Often seen in telephoto lenses



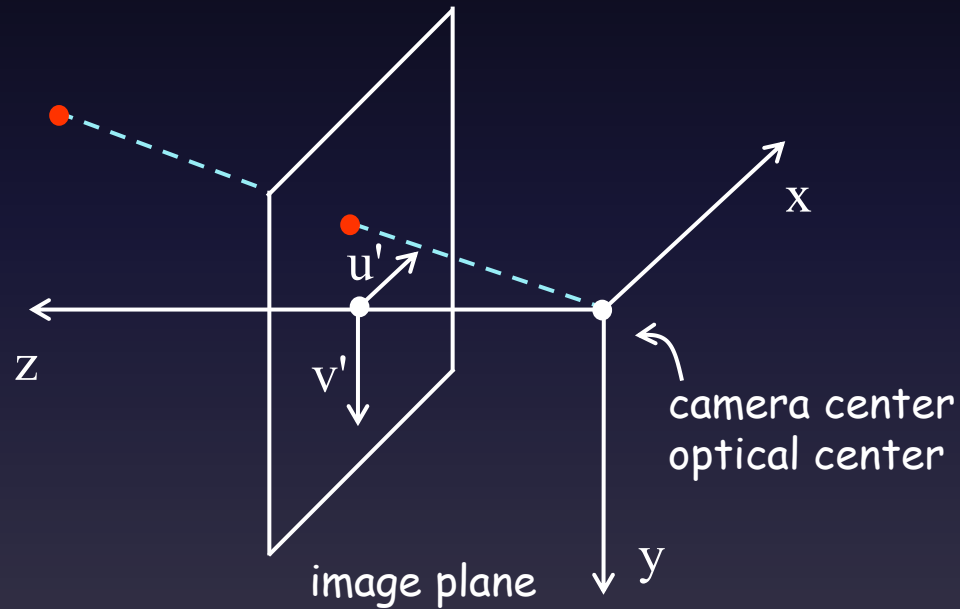
# Perspective Projection

- Pinhole Camera Model

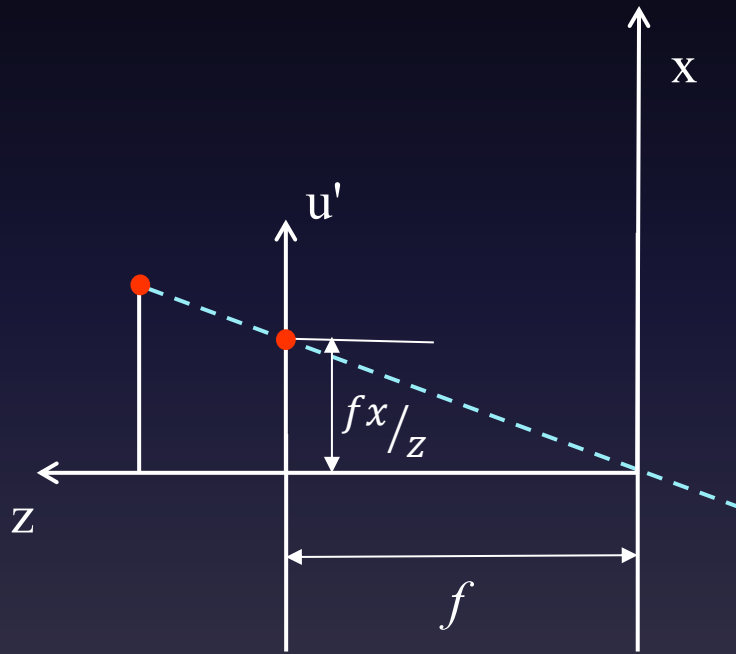


# Perspective Projection

- Pinhole Camera Model



# Perspective Projection



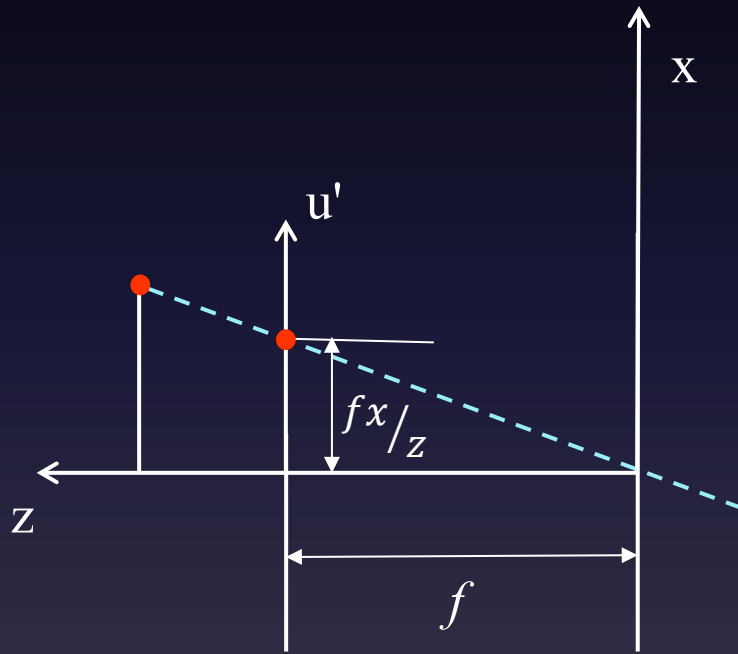
$$z : x = f : u'$$

$$u' = \frac{fx}{z}$$

Similarly,

$$v' = \frac{fy}{z}$$

# Perspective Projection



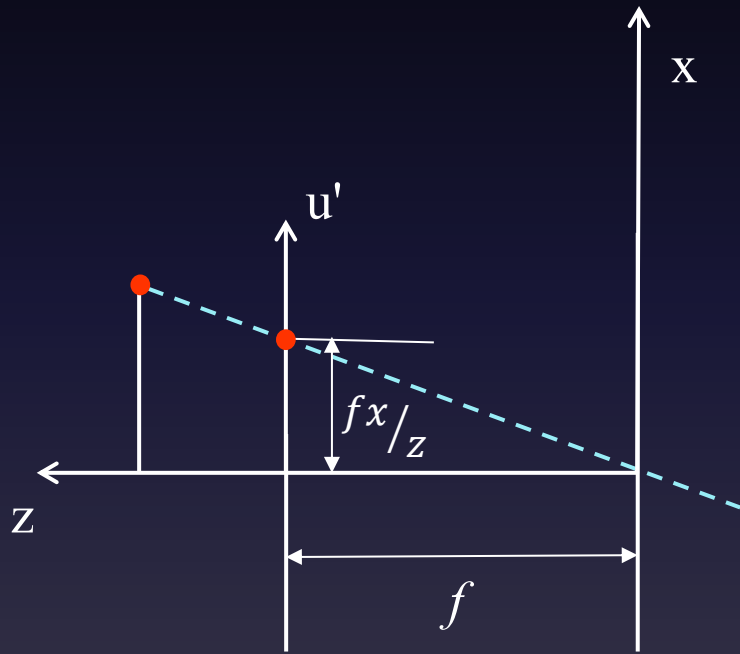
$$(x, y, z)^T \mapsto (u', v') = \left( \frac{fx}{z}, \frac{fy}{z} \right)$$

$$\begin{pmatrix} x \\ y \\ z \\ 1 \end{pmatrix} \mapsto \begin{pmatrix} U' \\ V' \\ S' \end{pmatrix} = \begin{pmatrix} fx \\ fy \\ z \end{pmatrix} = \begin{bmatrix} f & 0 & 0 & 0 \\ 0 & f & 0 & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix} \begin{pmatrix} x \\ y \\ z \\ 1 \end{pmatrix}$$

$$u' = \frac{U'}{S'}, \quad v' = \frac{V'}{S'}, \quad z = S'$$

linear projection in homogeneous coordinates !

# Perspective Projection



$$(x, y, z)^T \mapsto (u', v') = \left( \frac{fx}{z}, \frac{fy}{z} \right)$$

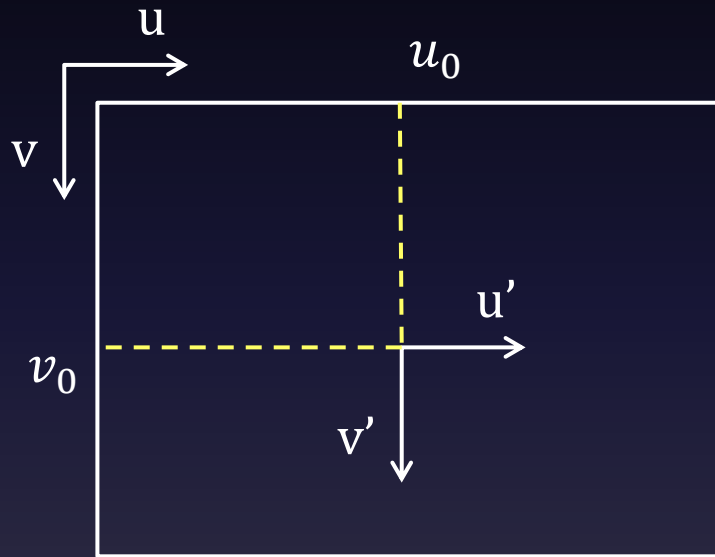
$$\begin{pmatrix} U' \\ V' \\ S' \end{pmatrix} = \begin{bmatrix} f & 0 & 0 \\ 0 & f & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix} \begin{pmatrix} x \\ y \\ z \\ 1 \end{pmatrix}$$

$$\tilde{\mathbf{U}} = \mathbf{P} \tilde{\mathbf{X}}_c$$

$$\mathbf{P} = \text{diag}(f, f, 1) [\mathbf{I} \mid \mathbf{0}]$$



# Principal Point



$(u_0, v_0)$

Principal point

$$(x, y, z)^T \mapsto (u', v') \mapsto (u, v)$$

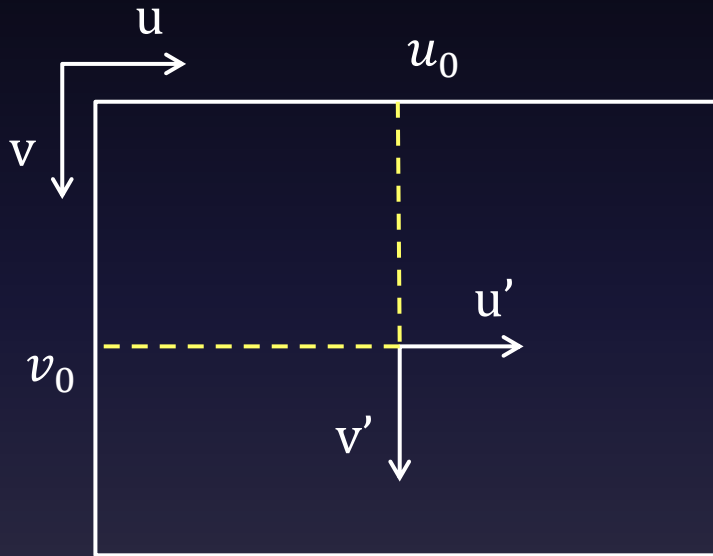
$$(u, v) = (u' + u_0, v' + v_0)$$

$$= (U'/S' + u_0, V'/S' + v_0)$$

$$\begin{pmatrix} U \\ V \\ S \end{pmatrix} = \begin{pmatrix} uS \\ vS \\ S \end{pmatrix} = \begin{pmatrix} (u' + u_0)S \\ (v' + v_0)S \\ S \end{pmatrix}$$

$$= \begin{bmatrix} f & 0 & u_0 \\ 0 & f & v_0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix} \begin{pmatrix} x \\ y \\ z \\ 1 \end{pmatrix}$$

# Principal Point



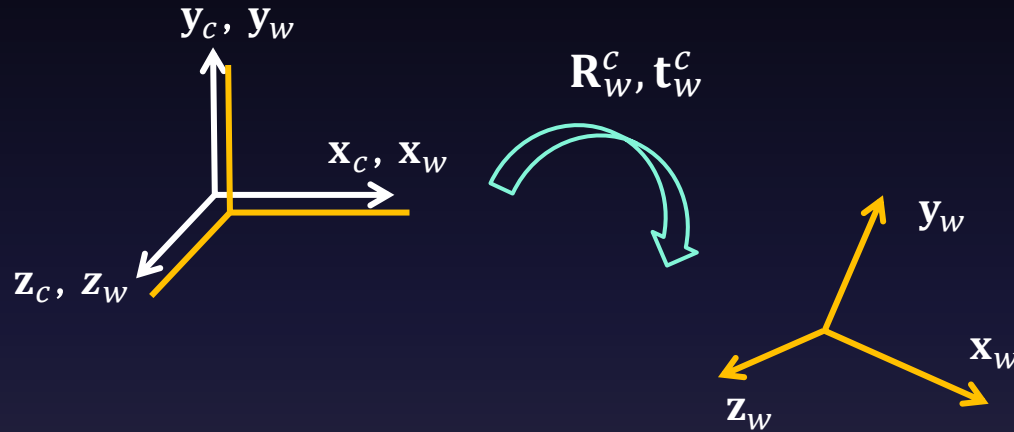
$$\begin{pmatrix} U \\ V \\ S \end{pmatrix} = \begin{bmatrix} f & 0 & u_0 \\ 0 & f & v_0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix} \begin{pmatrix} x \\ y \\ z \\ 1 \end{pmatrix}$$

$$\tilde{\mathbf{U}} = \mathbf{K} [\mathbf{I} \mid \mathbf{0}] \tilde{\mathbf{x}}_c$$

$$\mathbf{K} = \begin{bmatrix} f & 0 & u_0 \\ 0 & f & v_0 \\ 0 & 0 & 1 \end{bmatrix}$$

Camera matrix  
Intrinsic parameters

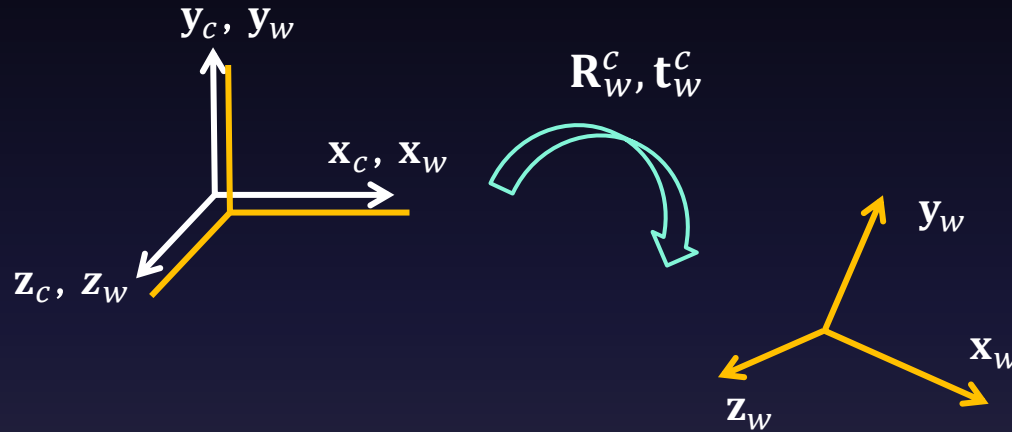
# Rotation and Translation



$$\mathbf{x}_c = \mathbf{R}_w^c \begin{pmatrix} x_w \\ y_w \\ z_w \end{pmatrix} + \mathbf{t}_w^c$$

$\mathbf{R}_w^c, \mathbf{t}_w^c$  : transform from the world coordinates to the camera coordinates

# Rotation and Translation



$$\tilde{x}_c = \begin{bmatrix} \mathbf{R} & \mathbf{t} \\ 0 & 1 \end{bmatrix} \begin{pmatrix} x_w \\ y_w \\ z_w \\ 1 \end{pmatrix} = \begin{bmatrix} \mathbf{R} & \mathbf{t} \\ 0 & 1 \end{bmatrix} \tilde{x}_w$$

Substituting into  $\tilde{\mathbf{U}} = \mathbf{K}[\mathbf{I} \mid \mathbf{0}] \tilde{x}_c$  gives  $\tilde{\mathbf{U}} = \mathbf{K}[\mathbf{R} \mid \mathbf{t}] \tilde{x}_w$ .

$$\tilde{\mathbf{U}} = \mathbf{P} \tilde{x}_w \quad \mathbf{P} = \mathbf{K}[\mathbf{R} \mid \mathbf{t}]$$

# Ideal Perspective Projection

$$\tilde{\mathbf{U}} = \mathbf{K} [\mathbf{R} \mathbf{t}] \tilde{\mathbf{x}}_w \quad \mathbf{K} = \begin{bmatrix} f & 0 & u_0 \\ 0 & f & v_0 \\ 0 & 0 & 1 \end{bmatrix}$$

[m], [mm]



[pixel]

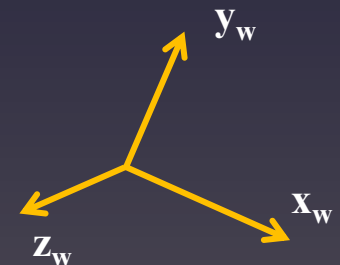
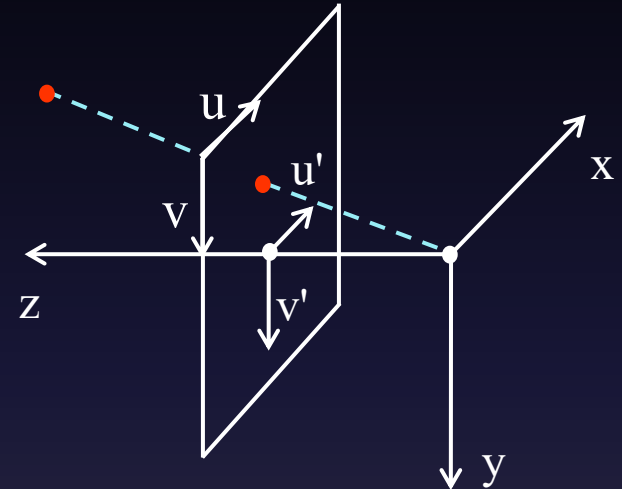
$$\mathbf{K} = \begin{bmatrix} k_u & 0 & 0 \\ 0 & k_v & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} f & 0 & u_0 \\ 0 & f & v_0 \\ 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} \alpha_u & 0 & k_u u_0 \\ 0 & \alpha_v & k_v v_0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$\equiv \begin{bmatrix} \alpha & 0 & u_0 \\ 0 & \beta & v_0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$\mathbf{P} = \mathbf{K} [\mathbf{R} \mid \mathbf{t}]$$

intrinsic camera parameters

extrinsic camera parameters



# Ideal Perspective Projection

$$\tilde{\mathbf{U}} = \begin{pmatrix} U \\ V \\ S \end{pmatrix} = s \begin{pmatrix} u \\ v \\ 1 \end{pmatrix} = \mathbf{K} \begin{bmatrix} \mathbf{R} & \mathbf{t} \end{bmatrix} \tilde{\mathbf{x}}_w$$

Fundamental Image Processing

Camera Calibration

